



**CSE3433 SOFTWARE ARCHITECTURE**

**PROGRAMME: COMPUTER SCIENCE ( SOFTWARE ENGINEERING)**

**SEMESTER II 2025/2026**

**PROJECT TITLE:**

**SMARTLEARN: CLOUD-NATIVE LEARNING MANAGEMENT SYSTEM (LMS)**

**LECTURER: DR. MOHAMAD NOR HASAN**

| <b>NO.</b> | <b>GROUP MEMBERS</b>                              | <b>MATRIC NO.</b> |
|------------|---------------------------------------------------|-------------------|
| <b>1.</b>  | <b>NURUL ALESSA BINTI MOHD NIZAM</b>              | <b>S76278</b>     |
| <b>2.</b>  | <b>NURAIN NAZIRAH BINTI ABDULLAH</b>              | <b>S75415</b>     |
| <b>3.</b>  | <b>PUTERI IREENE HAZWANI BINTI MUHAMAD HAIZAR</b> | <b>S75702</b>     |

## Weekly Progress

### Phase 1 (Week 3 – Week 5) – Project Proposal

| Week   | Activities Undertaken                                                        | Individual Contribution                                                                                                                       | Challenges                                                                    | Action Plan                                                                                   |
|--------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Week 3 | Initial discussion on SMARTLEARN LMS concept, objectives, and project scope. | Contributed backend development perspective by identifying system modules such as authentication, course management, and enrollment services. | Unclear project scope and uncertainty regarding required LMS functionalities. | Research existing LMS platforms and identify essential modules suitable for implementation.   |
| Week 4 | Requirement gathering and validation of system requirements.                 | Assisted in defining backend functional requirements, user roles, API requirements, and database interactions.                                | Difficulty narrowing down requirements due to multiple proposed features.     | Categorize requirements into essential and optional features to maintain project feasibility. |
| Week 5 | Proposal drafting and finalization.                                          | Prepared technical descriptions related to backend architecture, database structure, and security requirements.                               | Limited time available before proposal submission deadline.                   | Review proposal contents and verify consistency across all project sections.                  |

### Phase 2 (Week 6 – Week 7) – Architecture Design

| Week   | Activities Undertaken                                           | Individual Contribution                                                                                               | Challenges                                                           | Action Plan                                                               |
|--------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------|
| Week 6 | Designing system architecture and backend service interactions. | Proposed RESTful API architecture and designed communication flow between frontend, backend, and database components. | Limited experience with cloud-native architecture design concepts.   | Study architecture best practices and review similar LMS implementations. |
| Week 7 | Refining architecture diagrams and workflow processes.          | Mapped backend workflows for authentication, enrollment, course management, and grading operations.                   | Complex interactions between multiple user roles and system modules. | Simplify workflow diagrams and focus on major system functionalities.     |

### Phase 3 (Week 8 – Week 11) – Prototype Progress

| Week   | Activities Undertaken          | Individual Contribution         | Challenges                     | Action Plan                 |
|--------|--------------------------------|---------------------------------|--------------------------------|-----------------------------|
| Week 8 | Setting up backend development | Configured Node.js, Express.js, | Encountered package dependency | Verify all dependencies and |

|                |                                                              |                                                                                                                |                                                                                    |                                                                                    |
|----------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                | environment and database configuration.                      | MongoDB Atlas connection, and organized backend project structure.                                             | issues and database connection errors.                                             | establish stable database connectivity.                                            |
| <b>Week 9</b>  | Developing authentication and authorization functionalities. | Implemented user registration, login, JWT authentication, password hashing, and route protection middleware.   | Authentication tokens failed validation and login sessions were inconsistent.      | Improve token verification mechanisms and strengthen authentication security.      |
| <b>Week 10</b> | Developing business logic and API endpoints.                 | Created CRUD APIs for users, courses, enrollments, and announcements.                                          | API integration issues occurred when connecting frontend and backend modules.      | Test all endpoints using Postman and validate API responses before deployment.     |
| <b>Week 11</b> | Deployment preparation and backend integration testing.      | Configured environment variables, connected frontend (Vercel) with backend (Render), and tested deployed APIs. | Encountered CORS errors, failed API requests, and deployment configuration issues. | Update server configurations and replace localhost URLs with production endpoints. |

#### Phase 4 (Week 12 – Week 13) – Project Report, Individual Logbook & Final Demo

| <b>Week</b>    | <b>Activities Undertaken</b>                                           | <b>Individual Contribution</b>                                                                                     | <b>Challenges</b>                                                                                   | <b>Action Plan</b>                                                                     |
|----------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| <b>Week 12</b> | Conducting end-to-end testing and validating deployed services.        | Tested authentication flows, API functionality, database operations, and role-based access controls.               | Debugging production issues was difficult due to lengthy server logs and deployment-related errors. | Analyze logs systematically and perform comprehensive testing across all modules.      |
| <b>Week 13</b> | Preparing final report, individual logbook, and project demonstration. | Updated backend documentation, reviewed API specifications, and assisted in final system presentation preparation. | Ensuring all implemented features matched documented requirements and project objectives.           | Conduct final verification, resolve remaining issues, and complete project submission. |

## GenAI Usage Record

| Week    | Date       | Tool Used | Purpose                   | Prompt Used                                                                                                    |
|---------|------------|-----------|---------------------------|----------------------------------------------------------------------------------------------------------------|
| Week 8  | 11/05/2026 | ChatGPT   | Backend Project Structure | "Aku tengah buat LMS project. Untuk backend Node.js, folder structure yang kemas biasanya macam mana?"         |
| Week 8  | 13/05/2026 | ChatGPT   | Database Design           | "Untuk sistem LMS ni, collection MongoDB apa yang sesuai? Student, lecturer, course tu nak link macam mana?"   |
| Week 9  | 18/05/2026 | ChatGPT   | User Authentication       | "Kalau user login berjaya, JWT token ni sepatutnya simpan dekat mana supaya tak senang hilang?"                |
| Week 9  | 19/05/2026 | ChatGPT   | Password Encryption       | "Aku dah guna bcrypt, tapi macam mana nak compare password yang user masukkan dengan password dalam database?" |
| Week 9  | 21/05/2026 | ChatGPT   | API Endpoint Development  | "Boleh semak endpoint enroll course aku ni? Asyik return status code 400."                                     |
| Week 9  | 23/05/2026 | ChatGPT   | Backend Security          | "Untuk LMS system, route mana yang patut lecturer je boleh access dan route mana student boleh access?"        |
| Week 10 | 28/05/2026 | ChatGPT   | Git Repository Management | "Aku ter-push file .env ke GitHub. Apa risiko dia dan macam mana nak remove daripada repository?"              |
| Week 10 | 30/05/2026 | ChatGPT   | Render Deployment         | "Build dekat Render gagal masa deploy. Log dia tunjuk npm error. Nak start debug dari mana?"                   |
| Week 11 | 01/06/2026 | Gemini    | Environment Variables     | "Dekat Render ni environment variable tak detect. Ada setting yang aku terlepas ke?"                           |

|         |            |         |                         |                                                                                                                       |
|---------|------------|---------|-------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Week 11 | 02/06/2026 | ChatGPT | API Connectivity        | "Frontend dah deploy dekat Vercel tapi request API langsung tak sampai dekat backend Render."                         |
| Week 11 | 03/06/2026 | Gemini  | CORS Configuration      | "Origin blocked by CORS policy keluar dekat browser. Configuration Express aku ni ada yang salah ke?"                 |
| Week 11 | 04/06/2026 | Gemini  | Production Debugging    | "Kenapa API okay dekat localhost tapi bila production semua endpoint jadi error?"                                     |
| Week 11 | 06/06/2026 | ChatGPT | MongoDB Troubleshooting | "Database Atlas aku connected ke tak sebenarnya? Cara paling senang nak verify connection macam mana?"                |
| Week 12 | 08/06/2026 | ChatGPT | Form Submission Issue   | "User boleh submit form course, tapi data tak muncul dalam database. Nak trace masalah ni dekat bahagian mana?"       |
| Week 12 | 10/06/2026 | ChatGPT | Error Analysis          | "Ni stack trace daripada Render logs. Error ni sebenarnya datang daripada controller atau database query?"            |
| Week 13 | 15/06/2026 | ChatGPT | Report Writing          | "Boleh ringkaskan penerangan backend architecture SMARTLEARN supaya sesuai masuk dalam Software Architecture report?" |